

CCP REPORT

AI Sentinel: Smart Traffic Monitoring and Surveillance System

Data Mining (CT-356)



NED University of Engineering and Technology

Department of Computer Science and Information Technology

Submitted by:

Amna Altaf (DT-23003)

Sheikh Asma Javed (DT-23027)

Under the Guidance of:

Sir Rohail Qamar

Session: 2023 – 2027

INDEX

1. CCP Rubrics
2. Abstract
3. Introduction
 - 4.1 Background
 - 4.2 Problem Statement
 - 4.3 Objectives of the Project
4. Literature Review (2023–2025)
 - 5.1 YOLOv8 for Object Detection
 - 5.2 RT-DETR: Transformer-Based Detection
 - 5.3 Transformer Models (Hugging Face)
 - 5.4 Object Tracking using ByteTrack
 - 5.5 Intelligent Traffic Monitoring Systems
 - 5.6 AI-Based Weapon Detection
 - 5.7 Shoplifting Detection & Behavior Analysis
5. Methodology (Approach Used)
 - 6.1 System Overview
 - 6.2 Traffic Monitoring Module
 - 6.2.1 Object Detection
 - 6.2.2 Object Tracking
 - 6.2.3 Vehicle Counting
 - 6.2.4 Speed Estimation
 - 6.2.5 Alert Generation
 - 6.3 Surveillance Module
 - 6.3.1 Object Detection
 - 6.3.2 Weapon Detection
 - 6.3.3 Weapon-to-Person Assignment
 - 6.3.4 Shoplifting Detection
 - 6.4 Additional Techniques Used
 - 6.5 User Interface (Gradio)

6. Challenges in Vision-Based Speed Estimation

7.1 Perspective Distortion

7.2 Parallax Error

7.3 Impact on Speed Calculation

7.4 Solutions from Literature

7.4.1 Camera Calibration

7.4.2 Vanishing Point Method

7.4.3 Homography Transformation

7.4.4 Real-World Mapping

7.5 Limitations

7. Results and Discussion

8.1 Traffic Monitoring Results

8.2 Surveillance Results

8.3 Observations

8. Conclusion and Future Work

9.1 Conclusion

9.2 Future Work

9. References

RUBRICS

Criteria and Scales		
Satisfactory (2)	Average (1)	Unsatisfactory (0)
Criterion 1 Clarity of Objective: To what extent has the student understood the objective?_ (CP3: Depth of Knowledge Required)		
The student has correctly identified the requirements.	The student has correctly identified some requirements.	The student has misunderstood the requirements.
Criterion 2: Requirement Identification To what extent has the student understood the requirements? (CP9: Requirement Identification)		
The student has accurately identified and defined requirements	The student has partially identified and defined requirements	The student has wrongly identified and defined requirements
Criterion 3: Innovation: To what extent has the student used innovative analysis?(CP2: Depth of Analysis Required)		
The student has used conceptual thinking and innovative analysis to formulate suitable abstract models	The student has used some conceptual analysis to formulate suitable abstract models	The student has not performed thorough analysis and the model lacks required concepts.
Criterion 4: Modularity: To what extent has the student been able to identify the interdependent functionalities? (CP3: Depth of Knowledge Required)		
The student has applied advanced concepts with solid theoretical justification.	The student has applied upper-intermediate methods with good understanding.	The student has applied basic methods, limited theoretical understanding
Criterion 5: Validation: To what extent has the student rationalized the solution? (CP2: Depth of Analysis Required)		
The student has clearly justified that the solution meets the requirements.	The student has partially justified that the solution meets the requirements.	The student has not been able to justify that the solution meets the requirements.

Project Title:				
Group Members Roll #				
Total Marks:	_____/10		Teacher's Signature:	

Abstract

This project presents an intelligent Traffic Monitoring and Surveillance System using deep learning and computer vision techniques. The system is capable of detecting vehicles, monitoring traffic flow, identifying over-speeding vehicles, and performing surveillance tasks such as weapon detection and shoplifting detection.

For traffic analysis, the system uses object tracking and speed estimation to detect vehicles crossing a defined region and identify speed violations. For surveillance, it detects people, bags, and suspicious activities such as weapon carrying and item theft from shelves.

The system integrates multiple models including YOLO, RT-DETR, and a Hugging Face-based weapon detection model. A user-friendly interface is developed using Gradio to upload videos and visualize results.

The results show that the system can accurately detect objects, track movement, generate alerts, and provide useful insights for security and traffic management.

Introduction

With the increase in urban population, managing traffic and ensuring public safety has become a major challenge. Traditional monitoring systems rely heavily on human supervision, which can be slow and error-prone.

This project aims to develop an automated system that can:

- Monitor traffic in real-time
- Detect over-speeding vehicles
- Identify suspicious activities
- Detect weapons and theft incidents

The system combines Artificial Intelligence (AI) and Computer Vision to analyze video data and generate alerts automatically.

The main objectives of this project are:

- To build a smart traffic monitoring system
- To implement surveillance features like weapon detection

- To detect shoplifting behavior using object tracking
- To provide a simple interface for users

Literature Review (2023–2025)

Recent advancements in **computer vision and deep learning** have significantly improved intelligent traffic monitoring and surveillance systems. This section reviews key technologies and research contributions from the last three years.

1. YOLOv8 for Real-Time Object Detection (2023–2025)

YOLO (You Only Look Once) has remained one of the most widely used object detection algorithms due to its **real-time performance and high accuracy**. The latest version, YOLOv8, introduced by Ultralytics, improves both speed and detection precision compared to earlier versions.

Key Features:

- Single-stage detection (no separate region proposal step)
- High FPS (frames per second), suitable for real-time systems
- Improved detection of small and overlapping objects
- Anchor-free detection mechanism

Applications in Research:

Recent studies (2023–2025) show that YOLOv8 is widely used in:

- Traffic monitoring systems for vehicle detection
- Smart surveillance systems for identifying people and objects
- Autonomous driving applications

Relevance to Project:

In this project, YOLO is used for:

- Training a **custom weapon detection model**
- Real-time inference on video frames

Critical Analysis:

While YOLOv8 is fast and efficient, it may struggle in:

- Highly crowded environments
- Complex backgrounds

This limitation is addressed in this project by combining it with RT-DETR.

[\[1\]YOLOv8: A New State-of-the-Art Object Detection Model](#)

Link: <https://arxiv.org/abs/2304.00501>

[\[2\]You Only Look Once: Unified, Real-Time Object Detection](#)

Link: <https://arxiv.org/abs/1506.02640>

2. RT-DETR: Transformer-Based Real-Time Detection (2023)

RT-DETR (Real-Time Detection Transformer) is a modern object detection model that combines **transformer architecture** with real-time processing capabilities.

Key Features:

- End-to-end object detection
- Better global feature understanding
- Handles occlusion and dense scenes effectively
- No need for Non-Maximum Suppression (NMS) in some cases

Advantages Over YOLO:

- Better performance in crowded scenes
- More accurate object localization
- Strong contextual understanding

Applications:

- Smart city surveillance
- Crowd analysis
- Multi-object detection systems

Relevance to Project:

RT-DETR is used as the **main general detection model** in this system to:

- Detect persons, vehicles, and objects
- Provide input for tracking and behavior analysis

Critical Analysis:

Although RT-DETR improves accuracy, it:

- Requires more computational resources
- Is slightly slower than YOLO in some cases

[3]<https://arxiv.org/abs/2304.08069>

3. Transformer-Based Object Detection using Hugging Face (2023–2025)

Recent research highlights the growing use of **transformer-based architectures** in object detection tasks. Platforms like Hugging Face provide pre-trained models that can be easily integrated into applications.

Key Features:

- Pre-trained on large datasets
- Easy fine-tuning for custom tasks
- Supports transfer learning
- High accuracy in specialized detection

Applications:

- Weapon detection
- Medical imaging
- Custom object detection systems

Relevance to Project:

This project uses a **Hugging Face pre-trained weapon detection model**, which:

- Eliminates need for training from scratch
- Improves detection of weapons in real-world scenarios

Critical Analysis:

- May require GPU for efficient inference
- Performance depends on dataset used for training

4. Object Tracking using ByteTrack (2023–2024)

Object tracking is essential for analyzing motion, counting objects, and understanding behavior over time. ByteTrack is one of the most effective tracking algorithms used in modern systems.

Key Features:

- Assigns unique IDs to detected objects
- Maintains identity across frames
- Handles occlusion and missing detections
- Works with low-confidence detections

Applications:

- Traffic monitoring
- Surveillance systems

- Crowd tracking

Relevance to Project:

ByteTrack is used to:

- Track vehicles and people
- Enable speed calculation
- Prevent duplicate counting

Critical Analysis:

- Depends on detection quality
- Performance may degrade in very crowded scenes

[4\]ByteTrack: Multi-Object Tracking by Associating Every Detection Box](#)

Link: <https://arxiv.org/abs/2110.06864>

5. Intelligent Traffic Monitoring Systems (2023–2025)

Recent research focuses on developing automated traffic systems using AI to reduce manual monitoring.

Key Techniques:

- Line-crossing for vehicle counting
- Speed estimation using pixel displacement
- Region-based analysis
- Real-time alert systems

Applications:

- Smart cities
- Highway monitoring
- Accident prevention systems

Relevance to Project:

This project implements:

- Line-based counting
- Speed detection using frame-to-frame movement
- Speed alert generation

Critical Analysis:

- Speed estimation accuracy depends on camera calibration
- Performance affected by camera angle and resolution

[\[5\]End-to-End Object Detection with Transformers \(DETR\)](#)

Link: <https://arxiv.org/abs/2005.12872>

6. AI-Based Weapon Detection Systems (2023–2025)

Weapon detection has become an important area in surveillance research due to increasing security concerns.

Techniques Used:

- Convolutional Neural Networks (CNNs)
- Transformer-based models
- Hybrid deep learning approaches

Challenges:

- Detecting small objects (guns, knives)
- Variations in lighting and angle
- False positives in complex scenes

Relevance to Project:

This system:

- Uses a pre-trained model for weapon detection
- Assigns detected weapons to nearby persons
- Generates real-time alerts

Critical Analysis:

- Accuracy depends on dataset quality
- Real-time performance requires optimization

[\[6\]Automatic Gun Detection in Surveillance Videos Using Deep Learning](#)

Link: <https://ieeexplore.ieee.org/document/10054727>

[\[7\]Real-Time Weapon Detection in CCTV Using CNN](#)

Link: <https://arxiv.org/abs/2105.13097>

7. Shoplifting Detection and Behavior Analysis (2023–2025)

Modern surveillance systems are moving from simple object detection to **behavior analysis**.

Key Concepts:

- Tracking interaction between person and object
- Monitoring zones (e.g., shelf area)
- Detecting object disappearance

Techniques:

- Object tracking
- Temporal analysis (frame-by-frame changes)
- Proximity-based detection

Applications:

- Retail store monitoring

- Security systems
- Anomaly detection

Relevance to Project:

This project implements:

- Shelf zone detection
- Item tracking
- Theft detection when item disappears after interaction

Critical Analysis:

- May produce false alerts in crowded environments
- Requires consistent tracking for accuracy

[8][Video Anomaly Detection for Surveillance: A Survey](#)

Link: <https://arxiv.org/abs/2204.11176>

[9][Abnormal Event Detection in Surveillance Videos](#)

Link: <https://arxiv.org/abs/2001.04264>

Methodology

The proposed system is designed as a hybrid intelligent solution that combines deep learning-based object detection, tracking, and behavior analysis. It is divided into two major modules: **Traffic Monitoring Module** and **Surveillance Module**.

1. Traffic Monitoring Module

This module is responsible for analyzing traffic videos and extracting meaningful information such as vehicle count and speed.

- **Object Detection:**

The system uses the RT-DETR (Real-Time Detection Transformer) model to detect

different types of vehicles, including cars, buses, trucks, and motorbikes. RT-DETR is selected due to its high accuracy in complex and crowded environments.

- **Object Tracking:**

Detected vehicles are tracked across frames using the ByteTrack algorithm. This ensures that each vehicle is assigned a unique ID, enabling consistent tracking even in cases of partial occlusion.

- **Vehicle Counting:**

A line-crossing method is implemented where a virtual counting line is placed across the frame. Vehicles are counted when they cross this line, ensuring accurate counting without duplication.

- **Speed Estimation:**

Vehicle speed is calculated based on the displacement of object positions between consecutive frames. A pixel-to-real-world conversion factor is applied to estimate speed in km/h.

- **Alert Generation:**

If a vehicle exceeds the predefined speed limit (90 km/h), the system generates a real-time alert and highlights the vehicle in the output video.

2. Surveillance Module

The surveillance module focuses on security-related analysis and consists of three main components:

a) Object Detection

- The RT-DETR model is used to detect key objects such as persons, bags, and other relevant items.
- This serves as the foundation for further analysis such as weapon detection and theft detection.

b) Weapon Detection

- A pre-trained transformer-based model from Hugging Face is used for weapon detection.

- The model identifies potential weapons in video frames with high confidence.
- **Weapon-to-Person Assignment:**
The system calculates the distance between detected weapons and nearby persons. The weapon is assigned to the closest person.
- **Alert Generation:**
If a weapon is detected and associated with a person, the system generates an alert:
“Armed person detected”

c) Shoplifting Detection

- A predefined **shelf zone** is created within the video frame to monitor items.
- **Item Tracking:**
Objects such as bags or small items placed within the shelf zone are continuously tracked.
- **Disappearance Detection:**
If an item disappears for a certain number of frames, it is marked as missing.
- **Behavior Analysis:**
The system checks whether a person was near the item before it disappeared.
- **Alert Generation:**
If both conditions are satisfied, the system generates an alert:
“Theft detected”

Additional Techniques Used

- Bounding box visualization for detected objects
- Confidence threshold tuning to improve detection accuracy
- Detection fusion to remove duplicate bounding boxes
- Zone-based counting for controlled monitoring

- Frame-by-frame processing for real-time analysis

User Interface

The system interface is developed using Gradio to provide an easy and interactive experience.

- Users can upload a video file
- Select analysis mode:
 - Traffic
 - Surveillance
- The system generates:
 - Annotated output video
 - Detailed detection report

Challenges in Vision-Based Speed Estimation

Perspective Distortion Problem

In vision-based speed estimation systems, one of the major challenges is **perspective distortion**. In a 2D camera view, objects that are farther from the camera appear smaller and move less in pixel space, whereas objects closer to the camera appear larger and move more between frames.

This variation causes inconsistency when estimating speed using pixel displacement, as the same real-world movement produces different pixel movements depending on object distance.

Parallax Error

Another important factor affecting accuracy is **parallax error**.

Parallax error occurs due to differences in object depth relative to the camera. Objects at different distances are projected differently onto the image plane, which leads to measurement inaccuracies.

As a result:

- Vehicles far from the camera may appear slower
- Vehicles closer to the camera may appear faster

even if their actual speeds are identical. This makes raw pixel-based speed estimation unreliable without correction.

Impact on Speed Estimation

Speed estimation in computer vision systems is typically calculated as:

Speed = Change in position over time

However, in video analysis:

- Position is measured in **pixels**, not real-world units
- Pixels do not directly represent physical distance

Therefore, without proper calibration, any error in distance measurement directly affects the accuracy of speed estimation.

Solutions Proposed in Literature

Recent research in computer vision suggests several techniques to address these challenges:

1. Camera Calibration

Camera calibration is used to convert pixel measurements into real-world units (e.g., meters).

This is done by using known reference values such as:

- Road lane width
- Markings on the road

In this project, a pixel-to-meter conversion factor is applied based on approximate real-world measurements, which improves the accuracy of speed estimation.

2. Vanishing Point Estimation

Vanishing point methods use road geometry (such as lane lines) to estimate depth and perspective. This helps in understanding how objects scale with distance and allows better mapping between image space and real-world space.

3. Homography Transformation

Homography is a widely used technique that transforms the camera view into a **bird's-eye (top-down) view**.

This transformation:

- Removes perspective distortion
- Makes distances more uniform across the frame
- Improves accuracy of movement and speed estimation

4. Tracking with Real-World Mapping

Modern systems combine:

- Object detection
- Object tracking
- Real-world coordinate mapping

By tracking object movement across frames and converting pixel displacement into real-world distance, more accurate speed estimation can be achieved.

Limitations

Despite applying calibration and correction techniques, certain limitations remain:

- Camera angle can introduce distortion
- Depth variation across the scene affects scaling
- Approximate calibration may not fully match real-world measurements
- Environmental factors such as lighting and resolution impact detection accuracy

Results and Discussion

The system was evaluated using multiple video inputs, including traffic scenes and surveillance footage, to test its performance under different conditions.

Traffic Monitoring Results

- The system achieved accurate vehicle detection across different traffic scenarios
- The line-crossing method provided reliable and duplicate-free vehicle counting
- Speed estimation produced consistent and realistic results
- Speed alerts were generated correctly for vehicles exceeding the threshold

Surveillance Results

- The system successfully detected:
 - Persons
 - Bags and objects
 - Weapons
- The weapon-to-person assignment algorithm worked effectively, correctly associating detected weapons with nearby individuals
- The shoplifting detection module successfully identified cases where items disappeared after interaction with a person

Observations

- Combining multiple models (RT-DETR + transformer-based weapon detection + tracking) significantly improved system accuracy
- Real-time performance depends on hardware capabilities; GPU support greatly enhances processing speed
- Detection accuracy may vary depending on:

- Video resolution
- Lighting conditions
- Camera angle

Conclusion and Future Work

Conclusion

This project presents an intelligent and integrated system for traffic monitoring and surveillance using advanced deep learning techniques.

The system successfully demonstrates:

- Accurate object detection using modern models
- Reliable object tracking and counting
- Effective detection of suspicious activities such as weapon presence and shoplifting

Overall, the system provides a practical and scalable solution for smart city applications and security monitoring.

Future Work

The system can be further improved in the following ways:

- Enhance speed estimation using real-world calibration techniques
- Integrate face recognition for identity tracking
- Deploy the system on real-time CCTV networks
- Improve shoplifting detection using advanced behavior analysis models
- Optimize performance for real-time processing on low-end devices

References

1. **Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016).**
You Only Look Once: Unified, Real-Time Object Detection.
arXiv. <https://arxiv.org/abs/1506.02640>
2. **Ultralytics. (2023).**
YOLOv8: A New State-of-the-Art Object Detection Model.
arXiv. <https://arxiv.org/abs/2304.00501>
3. **Lv, W., et al. (2023).**
RT-DETR: Real-Time Detection Transformer.
arXiv. <https://arxiv.org/abs/2304.08069>
4. **Zhang, Y., Sun, P., Jiang, Y., Yu, D., Yuan, Z., Luo, P., Liu, W., & Wang, X. (2021).**
ByteTrack: Multi-Object Tracking by Associating Every Detection Box.
arXiv. <https://arxiv.org/abs/2110.06864>
5. **Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020).**
End-to-End Object Detection with Transformers (DETR).
arXiv. <https://arxiv.org/abs/2005.12872>
6. **(2023).**
Automatic Gun Detection in Surveillance Videos Using Deep Learning.
IEEE Xplore. <https://ieeexplore.ieee.org/document/10054727>
7. **(2021).**
Real-Time Weapon Detection in CCTV Using CNN.
arXiv. <https://arxiv.org/abs/2105.13097>
8. **Ramachandra, B., & Jones, M. (2022).**
Video Anomaly Detection for Surveillance: A Survey.
arXiv. <https://arxiv.org/abs/2204.11176>
9. **(2020).**
Abnormal Event Detection in Surveillance Videos.
arXiv. <https://arxiv.org/abs/2001.04264>